

BRENNER FIELD, NE, USAWMO: **725533**Lat: **40.080N**Lon: **95.592W**Elev: **980**StdP: **14.18**Time Zone: **-6.00 (NAC)**Period: **00-19**WBAN: **94957****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.3	4.0	-8.0	3.7	0.8	-4.1	4.6	5.6	26.6	34.5	24.4	31.7	6.7	320	0.536

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.1	95.3	77.2	92.4	76.1	89.8	75.1	80.1	90.4	78.4	88.7	76.9	86.8	11.4	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.2	146.8	86.7	75.4	138.0	85.0	73.8	130.5	83.2	44.4	90.6	42.6	88.8	41.0	86.6	85.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
24.5	20.9	18.8	DB	-8.1	100.6	4.8	3.1	-11.5	102.9	-14.3	104.7	-17.0	106.4	-20.4	108.7	
			WB	-8.7	83.2	4.6	1.3	-12.0	84.1	-14.7	84.9	-17.3	85.6	-20.6	86.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.5	27.2	29.6	43.0	54.3	64.5	74.1	77.7	75.1	67.9	55.1	42.1	30.3
	DBStd	20.29	12.04	12.17	11.88	9.98	8.47	6.00	5.68	5.65	8.06	9.60	10.50	11.22
	HDD50	2590	709	577	282	68	4	0	0	0	1	58	278	613
	HDD65	5474	1173	992	687	341	117	6	1	2	61	331	688	1075
	CDD50	3875	1	5	64	196	454	724	857	779	538	214	40	3
	CDD65	1284	0	0	4	19	101	280	393	316	148	23	0	0
	CDH74	13227	0	1	55	255	997	2855	4242	3083	1479	253	7	0
	CDH80	5371	0	0	8	63	313	1147	1932	1297	549	62	0	0
Wind	WSAvg	8.7	9.0	9.2	10.3	10.8	9.6	8.5	6.8	6.3	7.4	8.4	9.1	9.0
Precipitation	PrecAvg	32.8	0.7	1.1	1.9	3.2	4.6	5.0	4.9	3.5	2.9	2.5	1.8	1.0
	PrecMax	54.1	1.6	3.1	5.5	7.0	10.5	9.3	22.7	8.3	7.2	5.4	4.2	3.3
	PrecMin	21.5	0.0	0.0	0.0	0.9	0.8	1.1	1.3	0.7	0.1	0.2	0.0	0.0
	PrecStd	7.4	0.4	0.7	1.4	1.3	2.4	2.1	4.6	1.9	2.0	1.5	1.2	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.0	69.8	79.9	86.8	91.7	95.9	99.7	98.8	94.6	86.4	74.5	64.0
		MCWB	47.7	52.9	61.1	65.8	70.7	75.9	76.5	75.7	73.7	69.0	59.2	55.5
	2%	DB	55.1	60.0	73.9	80.7	86.9	92.4	95.9	93.8	90.1	80.6	69.1	56.3
		MCWB	44.3	48.5	57.9	62.6	69.2	75.2	77.9	77.2	73.3	64.6	57.0	47.9
	5%	DB	49.6	53.3	67.9	76.2	83.3	89.9	92.9	90.5	86.5	75.9	63.7	51.2
		MCWB	41.4	44.4	55.9	61.0	67.7	74.1	77.6	76.3	71.8	63.1	54.0	44.3
	10%	DB	43.9	47.7	62.4	71.8	79.9	86.8	89.8	87.1	82.5	71.7	59.1	46.6
		MCWB	38.2	40.8	53.1	59.3	66.5	73.1	76.6	74.4	68.8	60.9	51.2	40.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.7	55.3	64.6	68.4	74.4	80.0	82.9	82.1	77.4	72.5	62.3	57.8
		MCDB	54.6	63.0	74.6	81.3	85.2	91.2	91.6	92.0	89.2	82.0	69.6	61.4
	2%	WB	45.2	49.6	61.4	65.8	71.9	77.7	80.6	79.5	75.3	69.1	59.0	49.7
		MCDB	53.7	59.1	69.1	76.4	83.0	88.7	90.6	89.7	86.2	76.1	65.6	54.6
	5%	WB	41.9	45.0	58.1	63.1	70.1	75.9	79.1	77.8	73.5	65.3	55.5	44.7
		MCDB	49.5	53.1	66.7	73.6	80.7	86.5	89.4	87.6	83.1	72.5	62.8	50.4
	10%	WB	38.1	41.0	52.9	60.1	68.0	74.3	77.4	76.1	71.6	61.7	51.6	41.1
		MCDB	43.5	47.0	62.8	70.2	77.4	84.3	87.6	84.6	80.0	69.8	58.8	46.7
Mean Daily Temperature Range	5% DB	MDBR	19.4	19.7	21.7	22.6	21.0	20.3	20.1	20.7	23.1	22.5	21.3	18.4
		MCDBR	28.5	29.4	30.2	29.0	25.0	22.8	23.1	24.6	25.0	28.0	28.0	26.1
		MCWBR	19.5	19.2	17.3	15.8	11.5	10.0	10.2	11.2	11.0	14.6	17.0	18.4
	5% WB	MCDBR	26.7	27.6	24.6	24.8	21.6	20.3	20.3	20.6	22.1	22.0	24.3	23.7
		MCWBR	18.9	19.2	14.7	15.3	11.0	10.7	10.6	10.9	11.0	14.0	16.4	17.8
Clear-Sky Solar Irradiance	taub	0.280	0.304	0.329	0.380	0.411	0.420	0.425	0.424	0.379	0.335	0.299	0.284	
	taud	2.483	2.428	2.419	2.294	2.273	2.287	2.273	2.293	2.371	2.475	2.515	2.482	
	Ebn at Noon	284	291	294	284	276	272	270	267	273	276	275	273	
	Edh at Noon	24	30	33	40	42	42	42	40	35	28	23	23	
All-Sky Solar Radiation	RadAvg	682	955	1241	1571	1840	2058	2053	1812	1502	1068	751	579	
	RadStd	54	77	109	98	155	148	143	89	100	118	61	58	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (41 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page